





















































PROTOTYPE ONLY: added F101,U101..103,R101..105,C101..104,C202. Existing charger shown completely; parent integration pending.

F101 first at cell positive tab BEFORE ALL charger/boost branches. Cell tab-to-fuse conductor must be short, insulated and mechanically restrained.

U103 interrupts BQ input, including BQ IN-OUT short; not a cell-discharge switch. U101 senses actual cell tab independently of BQ feedback.

U102 holds ON low during monitor startup/brownout. OV fault opens U103; hysteretic retry, not a latched OVP. No firmware dependency.

PROPOSED charger tuning2.576V, same25mA: top22.2k/bottom10k both0.1%10ppm. No sibling edits. Healthy overshoot5mV allocation requires bench proof.

Fault-turnoff max absent; fuse I2t/resistance nominal only. Decaying fault clearing/temperature/voltage overshoot UNMET. Not all-single-fault safe.

Boundaries are external circuits/cell assembly, not invented physical connector pads. No layout fit or manufacturing-release claim.

Sheet: /
File: cell-protection.kicad_sch

Title: NOVA 25F cell OVP + cell-local fuse prototype

Size: A2 | Date: | **Rev: CP-1**

KiCad E.D.A. 10.0.7 | Id: 1/1